

Mason Davis

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EDUCATION

Master of Science in Computer Engineering

Expected June 2026

Utah State University, Logan, Utah

GPA: 4.0

- **Relevant Coursework:** Computer Architecture, Principles of Cyber-Physical Systems, Adv. Reconfig Architecture

Bachelor of Science in Computer Engineering, Minor in Computer Science

May 2024

Utah Valley University, Orem, Utah

Dean's List | GPA: 3.88

- **Certifications:** FAA Remote Pilot (2024), TensorFlow Developer (2023), Student Excellence Award (2023-24)

EXPERIENCE

Graduate Research Assistant | Utah State University

August 2024 - Present

- Developing VeRAPAK, a formal verification framework for neural network robustness.
- Formally modeling the Python-based VeRAPAK neural network framework in Dafny for computer-assisted proofs.
- Validating Python execution against Dafny models via a custom contract-constrained Atheris fuzzer.

Graduate Teaching Assistant | Utah State University

August 2024 - Present

- Guiding students in Dafny proofs and system modeling for Program Proofs & Cyber-Physical Systems courses.
- Mentored 15+ students in digital logic labs, troubleshooting Verilog implementations, and deployments on Basys 3.

Undergraduate Research Assistant | Utah Valley University

February 2023 - August 2024

- Led an ML team developing autonomous wind turbine inspection systems deployed on DJI Matrice and Mavic drones.
- Trained TensorFlow/Keras fault classification models and YOLO visual navigation models, achieving >90% accuracy.
- Architected custom computer vision data pipelines using OpenCV and CVAT for image augmentation and labeling.
- Engineered custom flight control software integrating real-time edge inference for GPS-denied autonomous flight.

Undergraduate Teaching Assistant | Utah Valley University

August 2023 - May 2024

- Designed and delivered technical lectures on Support Vector Machines, Deep Learning, and their applications.

ACADEMIC CONTRIBUTIONS

SPIN (2026), FMCAD (2025), IEEE i-ETC (2024)

Peer Reviewer

PUBLICATIONS

- **M. Davis** et al. "Identification and Localization of Wind Turbine Blade Faults Using Deep Learning" *Applied Sciences*, 2024.
- A. Rodriguez, **M. Davis** et al. "Deep Learning for Indoor Pedestal Fan Blade Inspection..." *Drones*, 2024.
- **M. Davis** et al. "Image Classification of Forest Fires Using Machine and Deep Learning." *IEEE i-ETC*, 2024.
- B. Altice, E. Nazario, **M. Davis** et al. "Anomaly Detection on Small Wind Turbine Blades..." *Energies*, 2024.
- **M. Davis**, M. Shekaramiz. "Desert/Forest Fire Detection Using Machine/Deep Learning Techniques." *Fire*, 2023.

PROJECTS

5-Stage DLX Processor | VHDL, Python, Assembly

- Implemented a 32-bit 5-stage pipelined custom DLX instruction set architecture in VHDL, verifying via Questa simulation.
- Engineered a Hazard Detection Unit with operand forwarding to minimize pipeline stalls and maximize throughput.
- Developed a custom Python assembler to compile DLX assembly into machine code.
- Added I/O support over UART with custom peripheral drivers and opcodes, synthesizing the final design onto a DE10-Lite.

Tetris | VHDL

- Designed and implemented a real-time Tetris-style game on a DE10-Lite using VHDL with a custom VGA controller.
- Verified system logic via Questa simulation and performed hardware analysis using the Signal Tap Logic Analyzer.

Embedded IR-Remote Device Driver | Verilog, C

- Engineered a hardware/software co-design for IR demodulation on a Zybo Z7 FPGA using a MicroBlaze processor.
- Integrated a custom Verilog IP core with a Linux-based embedded application via a custom C device driver.

SKILLS

Languages: Python, C, C++, VHDL, Verilog, Dafny, MATLAB/SIMULINK, LaTeX

Libraries: TensorFlow/Keras, OpenCV, PyTorch, scikit-learn

Tools: Git, Docker, Questa/ModelSim, Vivado, Quartus, UPPAAL, SpaceX, SPICE, MultiSim, Cadence (Virtuoso, SimVision)